EXPLORATORY DATA ANALYSIS

Cars Datasets 2025

Submitted by	Radha Krishan Tripathi
Institution	LNCT University
Project type	Exploratory Data Analysis (EDA)
Tools used	Python, Pandas, Matplotlib, Seaborn

A beginner-level study of dataset structure, data quality, distributions, and relationships between selected car attributes.

1. Objective and Dataset Overview

Objective. The objective of this project is to explore the Cars Datasets 2025 dataset, identify basic patterns in car specifications, check data quality, and communicate findings using descriptive statistics and visualizations. No machine-learning model is built as part of this EDA project.

Dataset structure

After removing duplicate rows, the dataset contains **1,214 rows**. The working notebook contains 14 columns because additional analysis columns were created during cleaning. The original dataset has 11 fields describing company, model, engine, capacity, horsepower, speed, performance, price, fuel type, seats, and torque.

Item	Description
Rows after duplicate removal	1,214
Original columns	11
Duplicate rows identified and removed	4
Additional working columns	Seats_Numeric and horsepower-related helper columns
Libraries	Pandas, NumPy, Matplotlib, Seaborn

The fields are mostly stored as text in the source file, including specifications that contain units or ranges. Numeric helper columns were created only where a simple numeric analysis was useful.

Note: The public/original URL for the CSV was not provided in the working materials. The source is therefore recorded as the dataset file supplied for this assignment; add the official URL if your instructor provided one.

2. Data Quality and Cleaning

The data was checked for duplicate records, missing values, inconsistent category labels, and values that could not be interpreted as simple numbers.

Cleaning steps performed

• **Duplicate records:** 4 duplicate rows were identified and removed, leaving 1,214 rows.
• **Missing values:** Missing values were checked using Pandas. The original data includes 3 missing values in CC/Battery Capacity, 6 in Performance, and 1 in Torque.
• **Fuel-type labels:** Similar spellings and labels were standardized, including “plug in hyrbrid” to “Plug-in Hybrid”, and equivalent Petrol/Hybrid or Petrol/Diesel labels were grouped.
• **Seat values:** A numeric helper column, Seats_Numeric, was created. Entries such as “2+2” and ranges such as “2–15” become missing in this helper column, while the original Seats column is retained.
• **Price field:** The entry “N/A (Concept Only)” was replaced with a missing value. Price ranges were retained because choosing one value would require an unsupported assumption.
• **Horsepower:** A numeric helper field was created by extracting the first number from each horsepower entry.

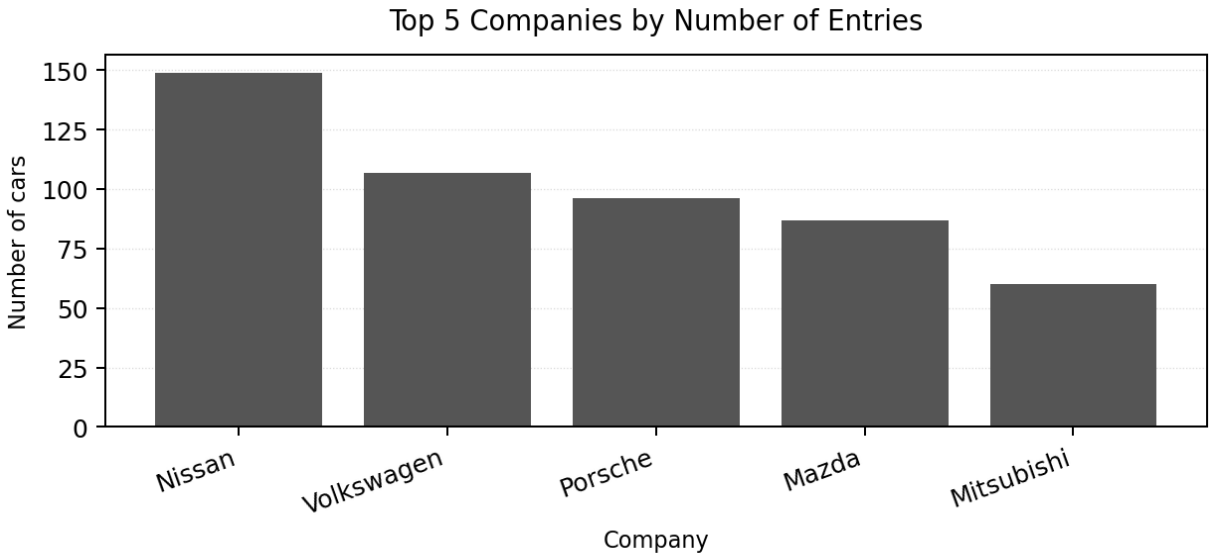
Missing-value summary after cleaning

Column	Missing values
CC/Battery Capacity	3
Performance(0 - 100)KM/H	6
Torque	1
Cars Prices	1
Seats_Numeric	12

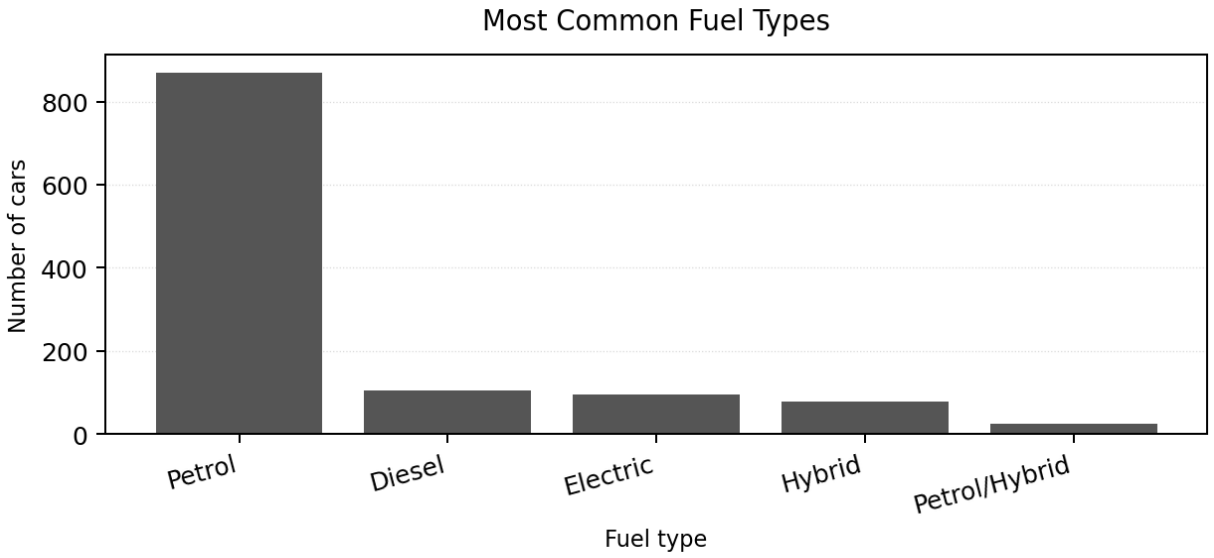
The 12 missing values in Seats_Numeric are expected helper-column values for non-numeric seat formats; they do not mean the original Seats entries were blank. Missing values were not filled with guessed values.

3. Descriptive Analysis and Visualizations

The following charts summarize the most common categories recorded in the dataset. Counts refer to entries in the cleaned dataset, not necessarily the complete global market.

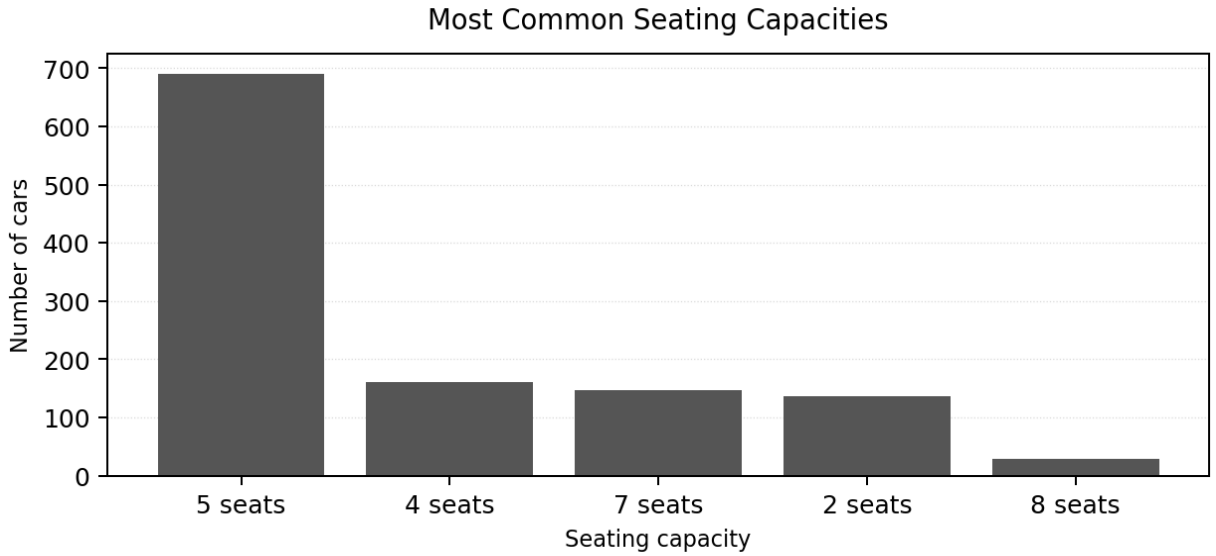


Observation: Nissan has the highest number of entries (149), followed by Volkswagen (107) and Porsche (96).



Observation: Petrol is the most frequent fuel label (869). Diesel (106), Electric (96), and Hybrid (78) appear much less frequently.

4. Seating Capacity and Horsepower



Observation: Five-seat cars are most common (690 entries), followed by four-seat (160) and seven-seat (146) cars. Some seat formats were excluded from numeric counts because they were written as ranges or as “2+2”.

Horsepower summary

The average of the HorsePower_Numeric helper column is approximately **300.8 hp** (about 301 hp). The helper column uses the first number from values such as “70-85 hp”, so the mean should be interpreted as an approximate summary rather than an exact horsepower measurement for every car.

Relationship between horsepower and seats

The correlation between HorsePower_Numeric and Seats_Numeric is approximately **-0.24**. This is a weak negative relationship in this dataset. Correlation alone does not show that seating capacity causes horsepower to change.

Other visualizations created in the notebook include a boxplot for seat-count outliers, a missing-values bar chart, a fuel-type-by-seating heatmap, and a scatter plot of horsepower versus seats. Keep these in the notebook and include the most relevant ones in the final report if space permits.

5. Key Findings

1. Company representation: Nissan has the largest number of entries (149), followed by Volkswagen (107), Porsche (96), Mazda (87), and Mitsubishi (60).
2. Fuel types: Petrol is the dominant fuel category with 869 entries. Diesel and Electric have 106 and 96 entries, respectively.
3. Seating capacity: Five-seat cars are the most common, with 690 entries. Four-seat and seven-seat cars follow with 160 and 146 entries.
4. Horsepower: The average extracted horsepower is about 301 hp. This figure is approximate because range values were represented by their first number.
5. Correlation: Horsepower and numeric seating capacity have a weak negative correlation (approximately -0.24), which should not be interpreted as causation.

6. Limitations

• Some specification fields contain missing values.
• Some prices are ranges; 168 entries were identified as containing a hyphen and were not reduced to a single price.
• Some seat values are ranges or formats such as “2+2”, so they cannot be used directly in numeric analysis.
• Extracting the first number from horsepower ranges simplifies the data and may understate the upper end of a range.
• The dataset may not represent all cars or all available models, and its original public source URL was not supplied in the working materials.

7. Conclusion

This exploratory analysis summarized the Cars Datasets 2025 file and highlighted patterns in company representation, fuel types, seating capacity, and horsepower. It also identified data-quality issues such as duplicate rows, missing values, inconsistent category labels, and ranges stored as text. The results provide a useful overview of the supplied dataset, while the limitations should be considered before drawing broader conclusions about the car market.

Dataset source: “Cars Datasets 2025.csv”, the file supplied for this assignment. Add the official dataset URL here if it was provided by your instructor.